

Dynamics and Learning on the IVM Lattice

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Contents

1 Dynamics and Learning on the IVM Lattice	1
1.1 Overview	1
1.2 Lattice sites, shells, and the 12-around-one move graph	1
1.3 Heat update and the monotonicity lemma	1
1.4 Majority update on integer states	2
1.5 Learning the coupling from an observed trajectory	2
1.6 Demo figure	3
1.7 Reproducibility and test contract	3
1.8 Cross-references	4

1 Dynamics and Learning on the IVM Lattice

1.1 Overview

Where [Section 4](#) descends a *single point* along the IVM lattice, this section evolves a *field* — one scalar per lattice site — and then asks the inverse question: given an observed field trajectory, what coupling produced it? Everything here is implemented in `ivm_dynamics.py` (`ball_sites`, `make_lattice`, `heat_step`, `majority_step`, `step`, `simulate`, `fit_trajectory`, `sum_of_squares`, `is_nonincreasing`) on top of the `Quadray` class and `to_xyz` embedding in `quadray.py`; the demo figure is produced by `quadmath/scripts/ivm_dynamics_demo.py` (which delegates to `render_dynamics_demo` in `src/quadmath/lattice/ivm_dynamics.py`, per the thin-orchestrator contract in `quadmath/scripts/AGENTS.md`).

1.2 Lattice sites, shells, and the 12-around-one move graph

A *site* is a canonical quadray representative — non-negative integer components with at least one zero, selected by `Quadray.normalize` (see [Quadray Methods](#)). The radial observable is the squared embedding norm `site_radius_sq(q)`; because the shift vector $(1, 1, 1, 1)$ lies in the kernel of the `DEFAULT_EMBEDDING` matrix, the radius is invariant under quadray normalization. `ball_sites(R)` enumerates the canonical sites with squared radius at most R^2 : for a canonical site the squared radius is at least twice the square of its largest component, so components in $[0, R]$ cover the ball.

`make_lattice(R)` joins two sites when one of the 12 canonical IVM neighbor shifts (`neighbor_shifts`, all permutations of $(2, 1, 1, 0)$), each at squared radius 8 — the close-packing distance) maps one site onto the other after renormalization. Normalization never moves the embedded point, so adjacency is exactly “embedding difference is one of the 12 shifts”. Lattice sites are the quadrays whose coordinate sum is divisible by 4; the remaining residue classes are IVM voids and are excluded. On the $R = 3$ ball (13 sites) the graph is a single connected component: the close-packed **12-around-one cluster**, i.e. the origin (degree 12) and its twelve cuboctahedron neighbors (degree 5). The update laws below act independently on each connected component of the ball.

1.3 Heat update and the monotonicity lemma

The heat (diffusion) update blends a field $u_t \in \mathbb{R}^N$ — one value per lattice site, N the site count — with its symmetric-normalized adjacency average $S = D^{-1/2}AD^{-1/2}$, where A is the adjacency matrix of the move graph, D the diagonal degree matrix, and rows of S are zero at isolated sites:

$$u_{t+1} = (1 - \alpha)u_t + \alpha S u_t, \quad \alpha \in [0, 1]. \quad (1)$$

Lemma (heat averaging is non-increasing in sum of squares). For every $\alpha \in [0, 1]$, every lattice built by `make_lattice`, and every state,

$$\|u_{t+1}\|_2^2 \leq \|u_t\|_2^2. \quad (2)$$

Proof sketch. S is symmetric with spectrum in $[-1, 1]$ (it is similar to the random-walk matrix $D^{-1}A$). The update operator $(1 - \alpha)I + \alpha S$ is therefore symmetric with eigenvalues in $[1 - 2\alpha, 1] \subseteq [-1, 1]$, hence non-expansive in the 2-norm. $\square$

This is exactly the claim the test suite asserts — *per step*, across seeds and the full coupling range (`test_heat_update_l2_nonincreasing_per_step` in `tests/unit/lattice/test_ivm_dynamics.py`). Two scope restrictions are deliberate and tested:

1. The guarantee is specific to the symmetric normalization S . Plain row-average diffusion $P = D^{-1}A$ on a truncated ball is **not** covered (boundary rows make P non-doubly-stochastic); the module neither uses nor claims it.
2. In the demo run ($\alpha = 0.35$, $T = 40$, seed 7) the sum of squares decreases monotonically from 37.04 to 3.93 while never increasing at any single step.

1.4 Majority update on integer states

For integer-valued fields the rounded-averaging (“majority”) update is

$$u_{t+1}(s) = \text{rint}\left((1 - \alpha)u_t(s) + \alpha \frac{1}{\deg(s)} \sum_{n \in N(s)} u_t(n)\right), \quad (3)$$

with $N(s)$ the neighbor set of site s and $\deg(s) = |N(s)|$ its degree, identity rows at isolated sites, and round-half-to-even ties (`numpy rint`), computed by `majority_step`. Every new value is a rounded convex combination of the site and its neighbors, so the extremal structure is preserved:

Proposition (range containment). Under (3) the field maximum is non-increasing and the minimum non-decreasing, at every step, for every $\alpha \in [0, 1]$.

We deliberately make **no** sum-of-squares claim for this update: rounding can increase it, and the test suite pins a demonstrating case (origin at 0 with its twelve packers at 1 maps to thirteen sites at value 1, raising the sum of squares from 12 to 13 — `test_majority_step_sum_of_squares_can_increase`). The honest summary: heat is L_2 -monotone, majority is range-preserving, and neither claim is stretched to cover the other update.

1.5 Learning the coupling from an observed trajectory

Given observed snapshots $u_{\text{obs}}(0..T)$ on a known lattice, the coupling α is identified gradient-free: `fit_trajectory` re-simulates the field from the observed initial condition for each candidate coupling and minimizes the trajectory mean squared error over the T post-initial snapshots and the N sites, where $u_\alpha(t)$ is the re-simulated field under candidate coupling α :

$$\mathcal{L}(\alpha) = \frac{1}{T N} \sum_{t=1}^T \|u_\alpha(t) - u_{\text{obs}}(t)\|_2^2, \quad (4)$$

scanning the supplied grid, then re-gridding the interval between the best candidate’s grid neighbors `refine_rounds` times (grid/coordinate search; ties resolve to the smallest α ; fully deterministic). The result is a global optimum *over the evaluated candidates only* — no convergence beyond the refinement resolution is claimed. In the demo configuration (synthetic trajectory generated at $\alpha_{\text{true}} = 0.3$ on the $R = 3$ lattice, $T = 40$), a 21-point grid plus refinement (33 MSE evaluations) recovers $\alpha \approx 0.300$ with $\mathcal{L} \approx 4.5 \times 10^{-33}$ — machine precision for a bitwise-deterministic re-simulation. When the true coupling lies off-grid, refinement converges to within a grid-spacing fraction of it (`test_fit_refinement_finds_offgrid_coupling`: 0.37 recovered to ≤ 0.0125 from a coarse $\{0, 0.25, 0.5, 0.75, 1\}$ grid).

1.6 Demo figure

Generated by `uv run python quadmath/scripts/ivm_dynamics_demo.py (MPLBACKEND=Agg)`; all randomness is seeded (`DynamicsParams.seed`), so the figure is reproducible byte-for-byte up to PNG encoding.

1.7 Reproducibility and test contract

- Tests: `PYTEST_DISABLE_PLUGIN_AUTOLOAD=1 uv run coverage run -m pytest tests/unit/lattice/test_ivm_dynamics.py -q` then `uv run coverage report` — 42 tests, 100% statement and branch coverage of `src/quadmath/lattice/ivm_dynamics.py`, no mocks, all examples real numerics with fixed seeds.
- Markdown: `uv run python quadmath/scripts/validate_markdown.py`.
- The two provable claims (lemma (2), the range proposition under (3)) are asserted per step; the non-claims (row-average diffusion, majority L_2 behavior) are pinned by the honesty tests described above rather than left implicit.

1.8 Cross-references

- Site geometry and normalization: [Quadray Methods](#).
- Discrete point descent on the same move set: [Optimization in 4D](#).
- Field/energy vocabulary (free energy, Fisher information): [Appendix B](#); equation index: [Appendix A](#).

IVM lattice dynamics: heat diffusion, majority rounding, coupling learning

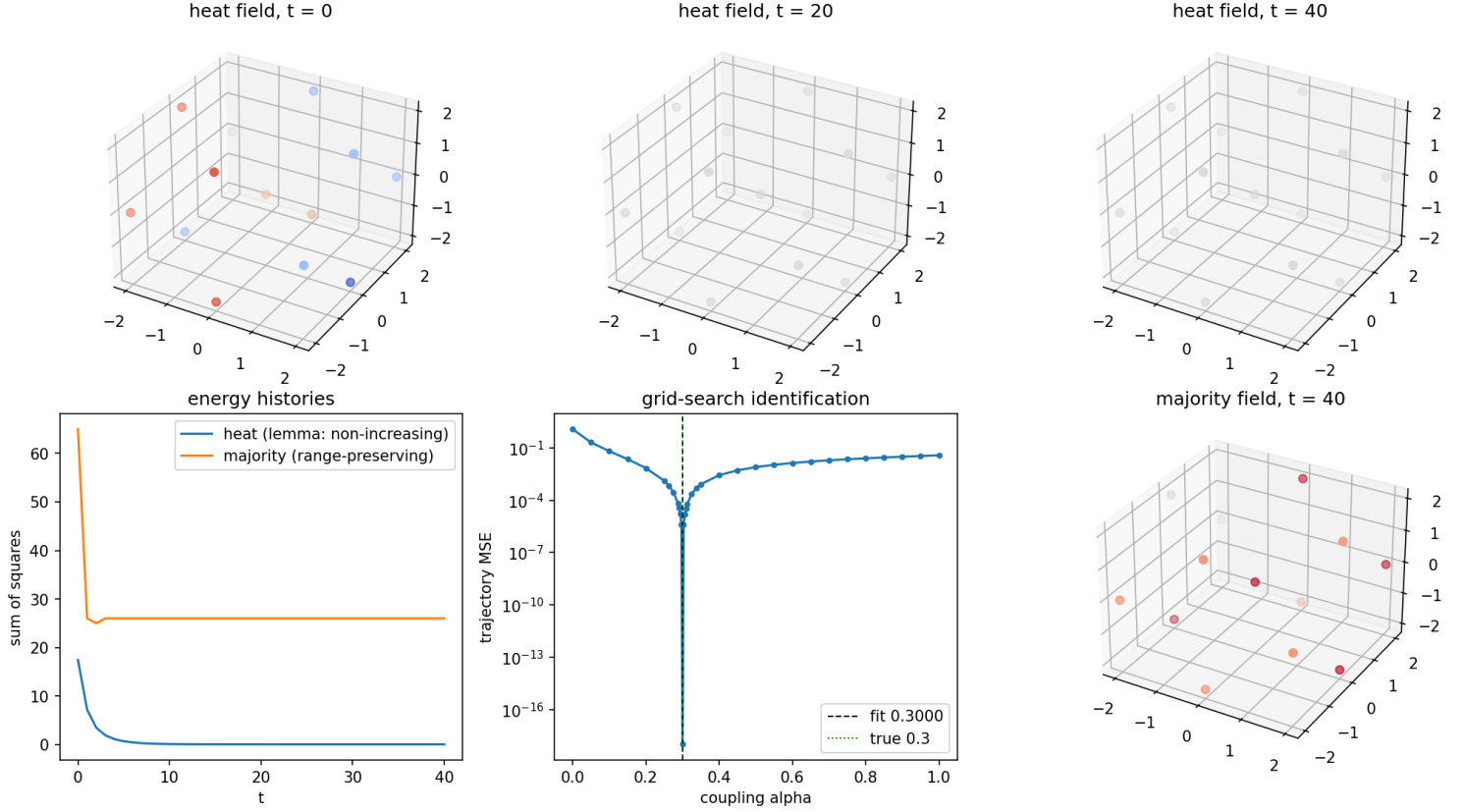


Figure 1: **Dynamics and learning on the IVM lattice ($R = 3$ ball, 13 sites)**. Top row: heat-field snapshots at $t = 0, 20, 40$ ($\alpha = 0.35$, seed 7) over the embedded quadray sites (XYZ axes in embedding units); dot color encodes field value on a shared symmetric scale, and diffusion homogenizes the field by $t = 40$. Bottom left: sum-of-squares histories over $t = 0 \dots 40$ — the heat curve decreases monotonically (lemma (2)); the majority curve falls steeply and plateaus, but unlike heat it carries no monotonicity guarantee, since rounding can increase the sum of squares (test_majority_step_sum_of_squares_can_increase). Bottom middle: trajectory MSE versus coupling α on a logarithmic axis over the search grid, with the identified optimum (fit 0.3000) coinciding with the true generating $\alpha = 0.3$ (dotted). Bottom right: the final integer majority field at $t = 40$. Generated by `uv run python quadmath/scripts/ivm_dynamics_demo.py (MPLBACKEND=Agg, seeded via DynamicsParams.seed)`.